L — Installation Guide

ZX Spectrum 128K, Commodore 64 and Atari 800XL / 130XE on the QMTECH Cyclone IV board: installing and updating the machines from an SD card. Release of 2 November 2026.

1. What is in this folder

File	What it is
`LUPDATE.BIN`	The machines: the machine menu, LSpectrum (ZX Spectrum 128K), L64 (Commodore 64), LTari 800 / 130 (Atari 800XL / 130XE), with the Spectrum and Atari ROMs. The only file you copy to the SD card.
`EN`, `SP`, `PT`, `DE`, `UA`, `RU`, `CN`, `KR`, `JP`	This guide and the user manual in nine languages.

Your board set comes ready: the ESP32 is programmed and the FPGA holds the machine menu. Only the machines have to be installed, from `LUPDATE.BIN`, and the same file is used for every later update.

2. What you need

- The board set: the QMTECH Cyclone IV core board with the adapter board, the ESP32 and the keyboard module.
- A USB keyboard, a VGA monitor and a 5 V power supply.
- A microSD card formatted **FAT32** or **FAT16**. **exFAT does not work**: most cards above 32 GB come as exFAT and must be formatted as FAT32 first (on Windows, for example with a FAT32 format tool; on a Mac with Disk Utility, "MS-DOS (FAT)").

3. Installing and updating

1. Copy `LUPDATE.BIN` from this folder to the **root folder** of the SD card (not into a sub-folder).
2. Put the card into the adapter board, connect monitor and keyboard, and switch the board on.
3. A new board shows **No machines installed. Copy LUPDATE.BIN to the SD card's root folder.** for a moment; a board that already has machines shows the machine menu. Within a few seconds the screen changes to **L update — Checking LUPDATE.BIN ...** and then **Writing the flash ...** with a progress bar.
4. **Do not switch off** while this is shown. The first installation takes about one minute; an update only rewrites what changed.
5. At the end the screen shows **Update complete.** and `LUPDATE.BIN` is deleted from the card. If it says **Switch the board off and on.**, do that.
6. The machine menu shows **1 LSpectrum, 2 L64, 3 LTari 800, 4 LTari 130**. Done.

If ...	Then ...
nothing happens	Check that the card is FAT32 or FAT16 and that `LUPDATE.BIN` is in the root folder (not in a sub-folder, not renamed).
the screen says **LUPDATE.BIN is not a valid** update or **Card read error.**	Nothing was written. Copy the file to the card again.
the power fails during the update	The file is still on the card: the update runs again at the next start and finishes the job.
the board shows nothing at all any more after a power failure during the update	Contact the seller of your board set: the machine menu has to be written again with a programming cable.
the screen says **(the flash was up to date)**	Nothing had to be changed.

> An update never touches the C64 ROMs you installed (section 4) and never your files on the SD card.

4. Software not included

The board set and this folder contain **no games and no other programs**, and **not the Commodore 64 ROMs**. Get them yourself and copy them to the SD card. Please respect the copyright: use only software and ROMs you are allowed to use; a game is not free just because it is old. Many old programs may be shared freely with their authors' permission, others may not.

Machine	Needs on the SD card	Where to look (examples)
LSpectrum	Nothing to start. Programs: .tap, .tzx, .z80	Spectrum Computing (spectrumcomputing.co.uk), World of Spectrum archives
L64	**The C64 ROMs (once, see below).** Programs: .d64, .prg, .t64, .tap	Use copies you are entitled to: read from a C64 and 1541 you own, or licensed ones (for example in Cloanto's C64 Forever). Programs: CSDb (csdb.dk), Commodore archives
LTari	Nothing to start. Programs: .atr, .xex, .car	Atarimania (atarimania.com), Atari archives

Installing the Commodore 64 ROMs (once)

1. On the SD card make the folder **/ROMS/C64/** and copy into it: the KERNAL (8 KB), BASIC (8 KB) and character ROM (4 KB) with names starting with **KERNAL**, **BASIC** and **CHAR**, and for the disk drive the 1541 DOS ROM (16 KB), name starting with **DOS1541** or **1541**.
2. Start **2 L64** in the machine menu. The ESP32 finds no ROMs in the flash, writes them there (a few seconds, shown on screen) and starts the C64.
3. From then on the C64 starts without the card. If the files are missing, the screen says where to put them: add them and start L64 again (Ctrl + Alt + Del, then 2). Without the 1541 file the

C64 runs, but has no disk drive.

```
SD card (example)
├── LUPDATE.BIN          (only while installing; deleted afterwards)
├── ROMS/
│   └── C64/
│       ├── kernal-901227-03.bin
│       ├── basic-901226-01.bin
│       ├── chargen-901225-01.bin
│       └── dos1541-325302-01+901229-05.bin
├── Spectrum/   (.tap .tzx .z80)
├── C64/        (.d64 .prg .t64 .tap .a64)
└── Atari/      (.atr .xex .car .a8s)
```

Folders may have any names; the file browser shows up to 320 entries per folder.

5. Quick check

Do this	You should see
Switch on	The machine menu with four machines; F3 opens the Bluetooth menu
Start **1 LSpectrum**, press **F12**	The 128K menu; F12 shows the files of the SD card
Ctrl + Alt + Del, start **2 L64**	The C64 READY screen (the first time: the ROMs are installed)
Ctrl + Alt + Del, start **4 LTari 130**	Altirra BASIC, READY
F8 in a machine	The picture switches between 60 Hz and 50 Hz

Everything else — keys, menus, joysticks, loading and saving — is described in the **User Manual** in the same folder.